NETWORK FILE SYSTEMS IN AWS

What is EFS

EFS is an elastic file system provided by amazon AWS that allows to connect the multiple EC2 instance to the same file system for shareable data.

IMPORTANT POINTS

Shared Storage

- Multiple EC2 instances can connect to the same EFS
- Useful for applications that need common data

Elastic (Auto Scaling)

- Storage size increases and decreases automatically
- No need to manually manage storage

Managed Service

- No need to manage disks or servers
- AWS handles everything

Uses NFS Protocol

- Works like a network drive in Linux
- Mount using NFS (Network File System)

Highly Available

- Data is stored across multiple Availability Zones

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What is elasticity and scalability

1. **Elasticity:** Ability of AWS to automatically adjust resources based on demand.
2. **Scalability:** Ability to increase or decrease resources to handle workload.

PRACTICAL-1 AMAZON LINUX

1.Create A Instance

1. name ---------> server1
2. ami ------------> amazon linux
3. key name ----> server1.pem
4. availability zone --> 1a
5. sg --------------> http/ssh/nfs (mysg)
6. create a instance

2.Create A Instance

1. name ---------> server2
2. ami ------------> amazon linux
3. key name ----> server2.pem
4. AZ ----------- --> 1b
5. sg --------------> http/ssh/nfs (mysg)
6. create a instance

3.Create A EFS Filesystem

1. name -------> EFS
2. click on -----> customize
3. File system type --> Regional
4. Next
5. Mount targets --> AZ-1a / AZ-1b / AZ-1c
6. Remove default security groups & add our mysg
7. Create a filesystem

4.Connect server-1

1. yum install amazon-efs-utils -y
2. mkdir efs
3. Go to EFS -> click on EFS -> Attach -> mount via DNS -> copy First URL
4. Paste the Copied URL
5. cd efs
6. touch file{1..10}
7. ls

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5.Connect Server-2

1. yum install amazon-efs-utils -y
2. mkdir efs
3. Go to EFS -> click on EFS -> Attach -> mount via DNS -> copy First URL
4. Paste the Copied URL
5. cd efs
6. ls
   {file1 file2 file3 file4 file5 file6 file7 file8 file9 file10}

PRACTICAL-2 UBUNTU

1.Create A Instance

7. name ---------> server1
8. ami ------------> amazon linux
9. key name ----> server1.pem

10. availability zone --> 1a
11. sg ----------> http/ssh/nfs (mysg)
12. create a instance

2.Create A Instance

7. name ---------> server2
8. ami ----------> amazon linux
9. key name ----> server2.pem
10. AZ ---------- --> 1a
11. sg ----------> http/ssh/nfs (mysg)
12. create a instance

3.Create A EFS Filesystem

8. name -------> EFS
9. click on -----> customize
10. File system type --> Regional
11. Next
12. Mount targets --> AZ-1a / AZ-1b / AZ-1c
13. Remove default security groups & add our mysg
14. Create a filesystem

4.Connect server-1

8. apt update
9. apt install nfs-common -y
10. mkdir efs
7. Go to EFS -> click on EFS -> Attach -> mount via DNS -> copy NFS client URL
11. Paste the Copied URL
12. cd efs
13. touch file{1..10}
14. ls

5.Connect Server-2

8. apt update
9. apt install nfs-common -y
10. mkdir efs
11. Go to EFS -> click on EFS -> Attach -> mount via DNS -> copy NFS client URL
12. Paste the Copied URL
13. cd efs
14. ls

    {file1 file2 file3 file4 file5 file6 file7 file8 file9 file10}

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